

Optimal Transport

Gautham Subramanian

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Haber, E., & Horesh, R. (2015). A multilevel method for the solution of Time Dependent Optimal Transport. Numerical Mathematics: Theory, Methods and Applications, 8(1), 97–111. <https://doi.org/10.4208/nmtma.2015.w02si>

Introduction

Consider two measures: a source measure μ_0 and a target measure μ_1 . The goal of optimal transport is to find a mass-preserving map T from μ_0 to μ_1 that minimizes some cost function. Below is a common formulation. Define $u = T(x) - x$ to be the displacement of each point. We want

$$\begin{aligned} \min \quad & M(u) := \frac{1}{p} \int_{\Omega} \mu_0(x) |u(x)|^p dx \\ \text{s.t.} \quad & c(u) = \det(I_d + \nabla u) \mu_1(x + u(x)) - \mu_0(x) = 0 \end{aligned}$$

This problem is famously nonlinear and difficult to solve, so we will express it in another way.

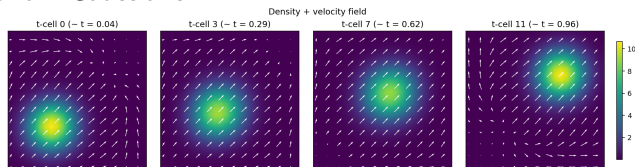
Rather than having a map from μ_0 directly to μ_1 , we will consider a time domain $t \in [0, 1]$ and a density function $\rho(x, t)$ such that $\rho(\cdot, 0) = \mu_0$, and $\rho(\cdot, 1) = \mu_1$; we also consider the "velocity" $v(x, t)$ rather than the displacement of each point:

$$\begin{aligned} \min \quad & \int_{t=0}^1 \int_{\Omega} \rho(x, t) |v(x, t)|^2 dx dt \\ \text{s.t.} \quad & \rho_t + \nabla \cdot (\rho v) = 0 \end{aligned}$$

We want thus want to find the optimal velocity function.

Optimal Transport

To get an idea of what the solution should look like, this is the solution when we have 2 Gaussians.



The goal of this paper is to solve this equation fast and efficient, and to improve upon the stability of previous papers (especially in cases that have high density contrast)

Consider the substitution $m = \rho v$:

$$\begin{aligned} \min \quad & \int_{t=0}^1 \int_{\Omega} \frac{|m(x, t)|^2}{\rho(x, t)} dx dt \\ \text{s.t.} \quad & \rho_t + \nabla \cdot m = 0 \end{aligned}$$

Notice that if we "combine" our spatial and time dimensions ($w = (x_1, \dots, x_m, t)^T$), our problem is effectively minimizing $f(w)$ subject to $\nabla^{\text{st}} \cdot w = 0$.

This has the same structure as of mixed form of non-linear flow in porous media, a problem that has been studied thoroughly. We can now use those techniques to solve our problem.

Discretization

The paper employed a finite-volume approach to discretizing the grid; we will do the same.

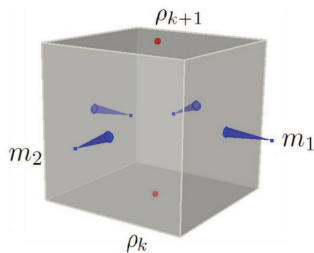


Figure 1: A box and the staggered grid in space-time. ρ is discretized at the beginning and end of each time interval and m is discretized on a staggered grid in space.

This approximation motivates us to use a staggered grid discretization.

We can incorporate boundary conditions into the problem if we write write:

$$q = h^{-1} \left(\mu_0^\top, \underbrace{0, \dots, 0}_{n_1 n_2 (n_3 - 1)}, \mu_1^\top \right)^\top = D_1 m_1 + D_2 m_2 + D_3 \rho$$

where D_j are derivative matrices in x , y , and t directions.

Additionally, we can show that

$$\int_{\Omega} \frac{|m|^2}{\rho} dx = h^3 (A_s m^2) A_t \left(\frac{1}{\rho} \right) + \mathcal{O}(h^2)$$

Where A_s and A_t are pointwise averaging matrices in space and time.

Note that in the previous slide, we wrote $A_s m^2$ and $A_t \left(\frac{1}{\rho} \right)$. The order in which we do the averaging matters a lot:

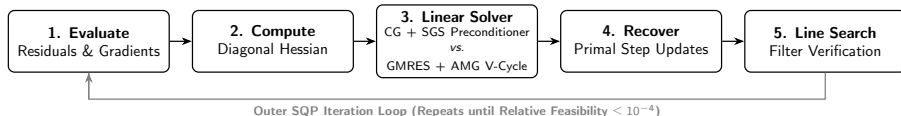
- We must square the mass then average (otherwise a sequence of rapidly alternating values 1, -1, 1, -1 gets mapped to zero spuriously).
- We must also invert then average to discourage the optimizer from approaching zero and negative values; this guarantees that $f(m, \rho) \rightarrow \infty$ as $\rho \rightarrow 0$.

Discrete Formulation + Optimization Problem

Discrete Version

$$\begin{aligned} \min_{m, \rho} \quad & h^3 \sum \left(A_s(|m|^p) \odot A_t(\rho^{-(p-1)}) \right) \\ \text{s.t.} \quad & D \begin{bmatrix} m^\top; \rho^\top \end{bmatrix}^\top - q = 0 \end{aligned}$$

We will use the inexact Newton Solver SQP:



The Lagrangian

We write the Lagrangian:

$$\mathcal{L}(m, \rho) = (A_s m^2) A_t \left(\frac{1}{\rho} \right) + \lambda^\top \left(D \begin{bmatrix} m^\top; \rho^\top \end{bmatrix}^\top - q \right),$$

and its first order necessary (KKT) conditions:

$$\nabla_m \mathcal{L} = 2MA_s^\top A_t \left(\frac{1}{\rho} \right) + (D_1, D_2)^\top \lambda = 0 \quad (1)$$

$$\nabla_\rho \mathcal{L} = -RA_t^\top A_s(m^2) + D_3^\top \lambda = 0 \quad (2)$$

$$\nabla_\lambda \mathcal{L} = D \begin{bmatrix} m^\top; \rho^\top \end{bmatrix}^\top - q = 0 \quad (3)$$

Even though this can be solved in theory, the cross terms in the Hessian make it very difficult; we want a Gauss Newton approximation of the Hessian by taking the diagonal $\hat{A} := \widehat{\nabla^2 \mathcal{L}}$. Note that this is SPD.

Solving the KKT System & Schur Complement

The full saddle-point KKT system is block-structured:

$$\begin{pmatrix} \hat{A} & D^\top \\ D & 0 \end{pmatrix} \begin{pmatrix} \delta w \\ \delta \lambda \end{pmatrix} = - \begin{pmatrix} \nabla_w \mathcal{L} \\ \nabla_\lambda \mathcal{L} \end{pmatrix}$$

We eliminate primal updates to form the symmetric positive definite (SPD) **Schur Complement**:

$$S \delta \lambda = \text{rhs}, \quad \text{where } S = D \hat{A}^{-1} D^\top$$

We solve this system (either by using GMRES + Ruge-Stuben style Algebraic Multigrid Preconditioner or with CG + Symmetric Gauss Seidel Preconditioner), and substitute to get the update weight to w :

$$\delta w = -\hat{A}^{-1}(D^\top \delta \lambda + \nabla_w \mathcal{L})$$

Fletcher-Leyffer Filter Line Search

We backtrack steps using $\alpha = (1, \frac{1}{2}, \frac{1}{4}, \dots)$, verifying three structural criteria:

- ➊ **Positivity Protection:** Ensure $\rho + \alpha\delta\rho > 0$ globally to prevent non-physical vacuums.
 - ➋ **Energy Control Cap:** Restrict sudden energy jumps via $h_{\text{trial}} \leq 1.3 \cdot h_{\text{current}}$.
 - ➌ **Filter Dominance:** The tuple (f, h) must not be dominated by prior entries within tolerances $\gamma_f = \gamma_h = 10^{-5}$.
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- **Staircase Prevention:** A trial state is only recorded in the filter history if $f_{\text{trial}} > f_{\text{current}}(1 + 10^{-4})$ to preserve search mobility.
 - **Restoration Fallback:** If all backtracks fail, the optimizer switches to a pure feasibility phase to minimize h directly.

Implementation Notes

- The paper considered what happens as $p \rightarrow 1$, but the ρ -block of $\hat{A}$ has $\rho^{-(p+1)}$ terms that blow up. I made sure that the elements of ρ were at least h^3
- I wanted to do as much as I could without constructing a matrix, but for the preconditioner must be constructed once per outer iteration to be passed into scipy/pyamg functions.
- I tuned the inner CG tolerance (opt= 5e-5) and the inner GMRES tolerance (opt= 0.025). These are both slightly lower than the paper's suggested values of 1e-4 and .1, but I found that reduction in SQP steps made up for the increase in the inner loop.
- I also implemented the filter for both CG and GMRES

Results 1

The paper first wanted to test the mesh-independence of the algorithm (since one of the primary benefits of SQP is that it is mesh-independent).

Table: Comparison Between Original Paper Results and Current Replication

Grid Size	Paper Results				Replication Results (Ours)			
	SQP	1 st	Midst	Last	SQP	1 st	Midst	Last
$16 \times 16 \times 10$	16	2	3	2	2	2	6	6
$32 \times 32 \times 20$	13	2	5	3	2	3	7	7
$64 \times 64 \times 40$	14	2	6	7	2	4	8	8

Analysis

Our results look pretty different because my GMRES tolerance is much higher than theirs, but I believe my tolerance lead to less work overall. However, both results do agree that this method is mesh independent insofar as the number of SQP iterations it takes.

Results 2

For the second experiment, the paper wanted to test the effectiveness of the algorithm with a very high density contrast with a multigrid approach:

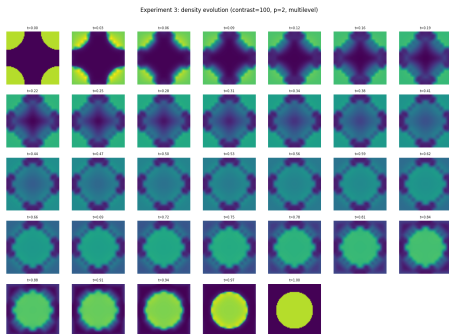


Figure: Evolution of OT problem with high density contrast

It produced a fairly stable solution and the paper's algorithm was a really robust way to solve this problem.

Results 3

This is the convergence of the previous slide is shown here:

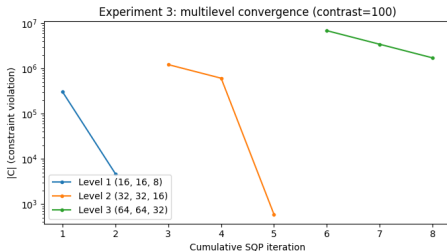
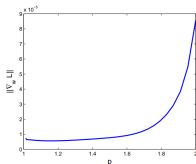
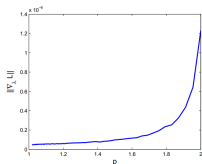


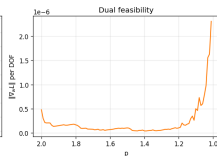
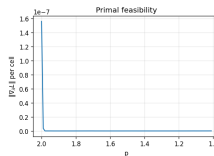
Figure: Multilevel Convergence of High Density Contrast Problem

Results 4

The final experiment was to consider what happened as $p \rightarrow 1$ (at $p = 1$, the problem is undefined, so they were trying to approximate it). This was done by iteratively lowering p by 0.01 between SQP iterations. I was not able to reproduce this.



Experiment 5: p-continuation (1 SQP step per p)



Paper's primal and dual feasibility

My primal and dual feasibility

I believe there is a stability issue with my discretization of the problem or maybe with my preconditioner, because these results refute the paper's claim that we can generate an approximation for the L1 problem.

I additionally considered 3 more types of problems:

- An asymmetric problem with two Gaussian that had different STDs
- Transforming a ring into a disk
- Transforming two blobs into one

Although I had some stability issues with the first two, the last one converged well. Since these are gifs, I'll discuss their results outside of the slideshow.